Elective 3

SYSTEM AND NETWORK ADMINISTRATION III

LEARNING OBJECTIVES

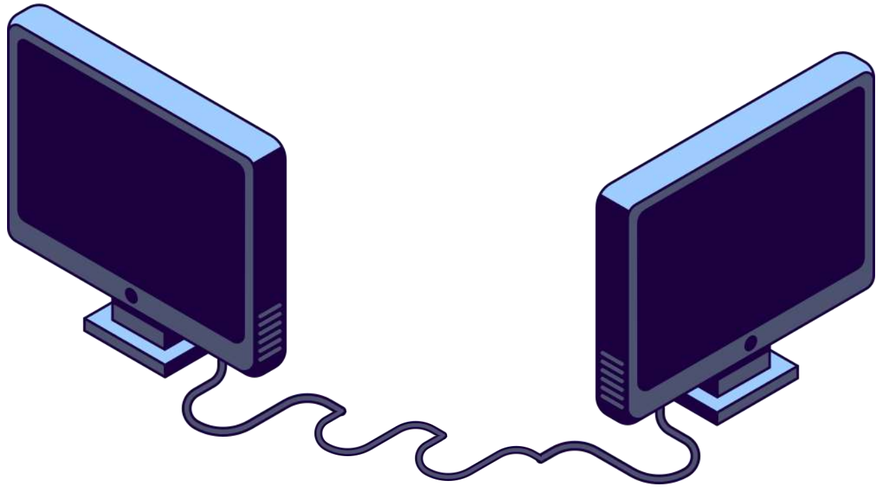
Understand core system administration concepts.

Develop network administration skills.

Apply security and cloud computing best practices.

COMPUTER NETWORKS

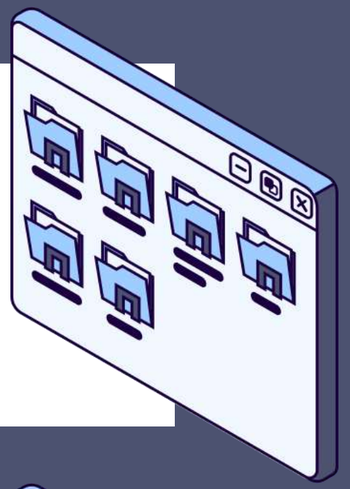
A computer network is two or more computers connected together, whereas a computer that is not connected to a network is stand-alone.



The term ‘computer’ can mean a range of different devices, including laptops, tablets and smartphones.

ADVANTAGES OF NETWORKING

Files are stored centrally on a server so can be accessed on computers across the network.



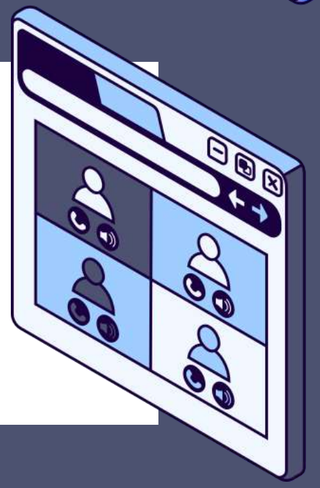
One internet connection can be used for multiple different devices using a router.



Devices (e.g. printers, scanners) are accessible on any network connected computer.

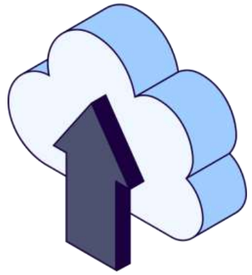


User accounts and software can be deployed across network computers.



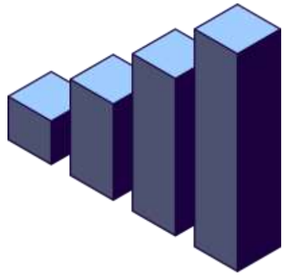
DISADVANTAGES OF NETWORKING

Availability



If connection is lost, files and devices are unavailable.

Performance



Larger networks are slower and less reliable.

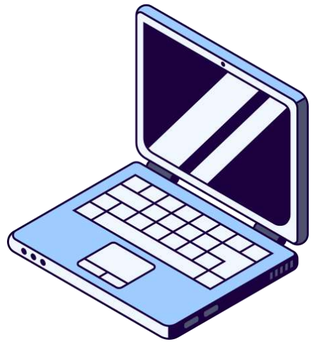
Management



Specialist knowledge and skills are required.

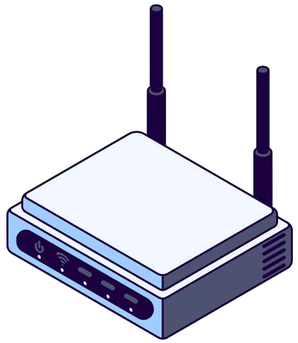
TYPES OF NETWORKS

PAN



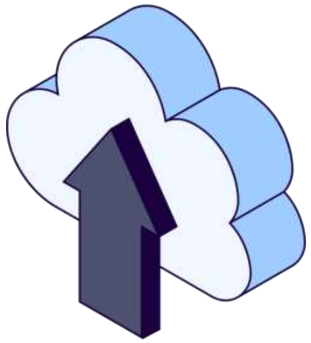
Personal Area Network

LAN



Local Area Network

WAN



Wide Area Network

PERSONAL AREA NETWORK

PAN (Personal Area Network) is used within a very small area. This is usually on a person or in a vehicle.

Bluetooth is usually used to connect these devices, which has a typical range of about 10 metres.

Examples of PANs:

- smartwatches
- wireless headphones
- wireless keyboards/mice
- car entertainment systems



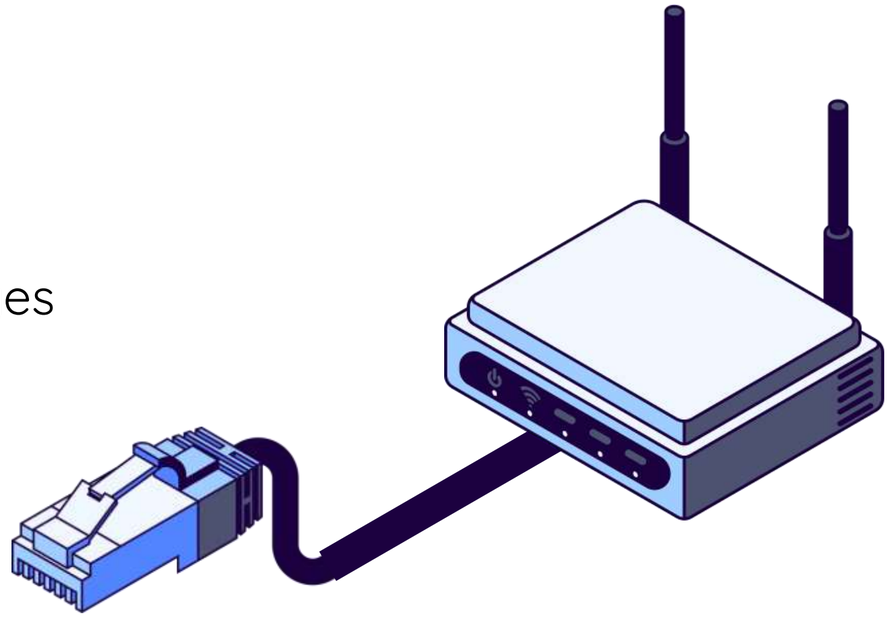
LOCAL AREA NETWORK

LAN (Local Area Network) is used within a small geographical area, which is usually limited to one building.

These networks can be connected wirelessly but for reliability, wired hardware (Ethernet) is often used.

Examples of LANs:

- home networks
- small businesses
- schools and colleges



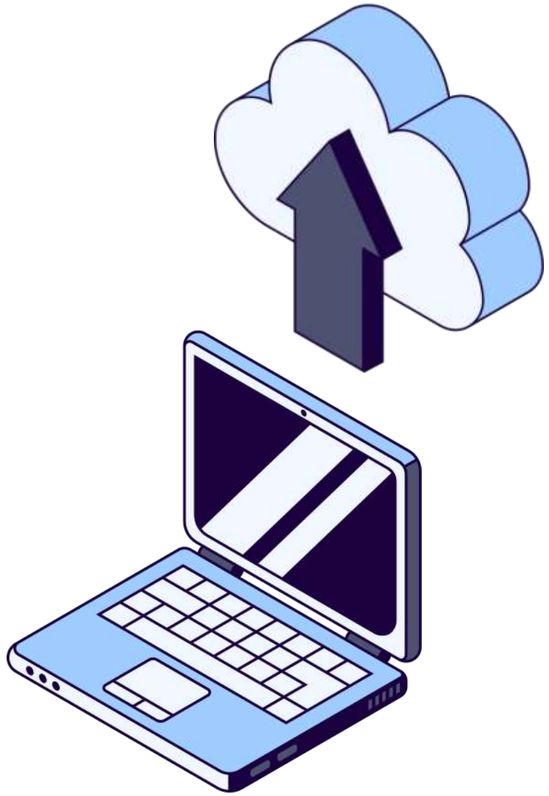
WIDE AREA NETWORK

WAN (Wide Area Network) is spread across a large geographical area, often an entire country or further.

The internet is a large scale WAN, which uses undersea cables to connect computers across the world.

Examples of WANs:

- the internet
- mobile networks
- banking networks



Drag and drop the network examples into the correct category.

Smartwatch	Office network	Bank network
Home network	In-car systems	Headphones
School network	The internet	Mobile network

PAN	
LAN	
WAN	

Drag and drop the network examples into the correct category.

PAN	Smartwatch	Headphones	In-car systems
LAN	Home network	Office network	School network
WAN	The internet	Mobile network	Bank network

SYSTEM ADMINISTRATION

Operating System (OS) – Software that manages hardware and software resources on a computer. Examples: Windows, Linux, macOS.

Kernel – The core component of an operating system that interacts directly with hardware.

Process Management – The handling of active processes in a system, including scheduling and execution.

File System – The structure used by an operating system to store and manage files. Examples: NTFS, FAT32, ext4.



User Account – A profile with permissions to access a computer or network.

Root (Superuser) – The highest privilege level in Unix/Linux systems that allows full control over the system.

Virtualization – The creation of virtual versions of hardware, such as servers or operating systems, using software like VMware or Hyper-V.

Patch Management – The process of updating software and operating systems to fix security vulnerabilities.

Backup & Recovery – The process of creating and restoring copies of data in case of system failures.

NETWORK ADMINISTRATION

IP Address (Internet Protocol Address) – A unique identifier assigned to each device on a network. Example: IPv4 (192.168.1.1), IPv6 (::1).

Subnetting – Dividing a network into smaller segments to improve performance and security.

DNS (Domain Name System) – A system that translates domain names (e.g., google.com) into IP addresses.

DHCP (Dynamic Host Configuration Protocol) – A protocol that automatically assigns IP addresses to devices on a network.



MAC Address (Media Access Control Address) – A unique identifier assigned to a network interface card (NIC).

Firewall – A security system that monitors and controls incoming and outgoing network traffic.

VPN (Virtual Private Network) – A secure connection that encrypts internet traffic to protect privacy.

Proxy Server – An intermediary between users and the internet that enhances security and anonymity.

Bandwidth – The amount of data that can be transmitted over a network in a given time.

Latency – The delay between sending and receiving data across a network.

Ping – A command used to check network connectivity between devices.

Traceroute – A tool used to map the path that data takes to reach its destination.

QoS (Quality of Service) – A networking feature that prioritizes certain types of traffic for better performance.

Security & Authentication

Authentication – The process of verifying a user's identity (e.g., password, biometrics, two-factor authentication).

Authorization – The process of granting a user specific permissions to access resources.

LDAP (Lightweight Directory Access Protocol) – A protocol used for managing user authentication in networks.

Active Directory (AD) – A Microsoft service that manages users, computers, and permissions in a network.

Multi-Factor Authentication (MFA) – A security process requiring two or more forms of verification.

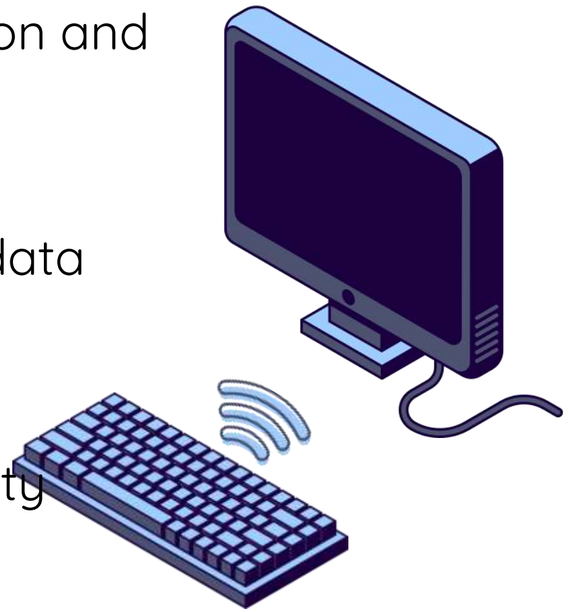
Encryption – The process of converting data into a secure format that cannot be easily read by unauthorized users.

SSL/TLS (Secure Sockets Layer / Transport Layer Security) – Protocols that encrypt data transmission over the internet.



COMPLIANCE & IT STANDARDS

- ISO 27001 – A global standard for information security management systems (ISMS).
- ITIL (Information Technology Infrastructure Library) – A set of best practices for IT service management (ITSM).
- GDPR (General Data Protection Regulation) – A European Union regulation on data protection and privacy.
- SOC 2 (Service Organization Control 2) – A framework for assessing an organization's data security practices.
- NIST (National Institute of Standards and Technology) – A U.S. agency that sets security standards for IT systems.



AUTOMATION & SCRIPTING

- Scripting – Writing small programs to automate repetitive tasks. Examples: Bash, PowerShell, Python.
- Cron Job – A scheduled task in Unix/Linux systems that runs automatically at specified intervals.
- PowerShell – A task automation framework used in Windows environments.
- Ansible – An IT automation tool used for configuration management and deployment.
- CI/CD (Continuous Integration/Continuous Deployment) – Automation in software development to streamline testing and deployment.
- DevOps – A methodology that integrates software development and IT operations to improve efficiency.